

Derleen Saini

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HIGHLIGHTS

- Solid data and analytics foundation, including SQL querying, data cleaning, ETL, business intelligence dashboarding and end to end analytics product development.
- Strong background in NLP and generative AI for business decisions including sentiment analysis pipelines and LLM benchmarking demonstrated in internships and graduate level coursework.

EDUCATION

M.S. Information Systems, Santa Clara University

September 2023-June 2025

B.S. Management Information Systems and Business Analytics, Santa Clara University

September 2020-March 2024

WORK EXPERIENCE

Data Analyst Intern, Microsoft (MSIS Practicum)

January 2025-June 2025

- Benchmarked 9 LLM and NLP tools (Copilot, GPT-4o, Gemini, Azure AI, FinBERT) on 4,800+ labeled financial sentences, delivering an accuracy, latency, and usability evaluation framework that informed Microsoft's recommended analytics stack and surfaced 3 optimization opportunities for Copilot integration.
- Productionized a multi-model sentiment analysis pipeline on 8+ quarters of Microsoft earnings call transcripts (100K+ words), segmenting results by business line and applying SHAP interpretation to uncover sentiment-to-stock correlations, delivering actionable insights to cross-functional finance stakeholders.
- Co-authored a published whitepaper (Microsoft Tech Blog, May 2025) and prototyped a Streamlit chatbot with GPT-4o for stakeholder self-service querying, strengthening Microsoft's thought leadership in AI-powered analytics.

Consumer Insight Data Analytics Externship, Beats by Dre

August 2024-September 2024

- Delivered consumer insights dashboard analyzing 1,000+ Amazon reviews (Python, NLP, Gemini AI), surfacing 10 high-impact themes that informed marketing strategy, contributing to a 10% lift in targeting effectiveness and a 5% gain in engagement/satisfaction.
- Presented data-driven KPIs and segmentation insights to cross-functional marketing stakeholders, shaping customer targeting and campaign strategy decisions.

Data Analyst Intern, Distribind

June 2022-August 2022

- Customized and transformed client datasets in Excel (Pivot Tables, XLOOKUPS, advanced data functions) to integrate 7 enterprise clients into Distribind's platform, streamlining stakeholder reporting workflows and improving operational efficiency by 5%.
- Validated reporting accuracy and optimized data structures, resolving critical errors that reduced processing failures by 34% and increased trust in business intelligence outputs.
- Launched the U.S. client analytics platform, enabling stakeholders to monitor market trends and driving a 20% reduction in portfolio risk exposure through data-informed policy recommendations.

PROJECTS

Data Warehousing & Business Intelligence Project – Divvy Bike Share System | Python, ETL, EDA, SQL, Tableau

- Engineered a full analytics lifecycle for the Divvy bike share system (21M+ trip records), from ETL into a MySQL star schema to interactive Tableau dashboards, driving recommendations on pricing, station rebalancing, and membership growth.

Generative AI to Create Stories | Python – Streamlit, NLTK, LangChain, Open AI – GPT Models, Dall-e, Text to Speech

- Created an interactive AI storytelling product, fine-tuning LLMs to generate text, illustrations and text-to-speech, demonstrating applications of generative AI in digital media.

Predicting Stock Price Movement | SpaCy, Document-Term matrix, Random Forest, Logistic Regression, AutoGluon, SVM

- Developed and validated machine learning models as predictive data products, achieving F1 score of 0.80 on 40 healthcare earnings call transcripts demonstrating reliable predictive accuracy for financial decision-making.

SKILLS

- **BI/Analytics/ML:** Tableau, Power BI, KPI Design, Natural Language Processing, Machine Learning (Classification, Regression, Clustering, Generative AI), Data Warehousing (ETL), Claude, GPT, Copilot, Prompt Engineering
- **Tools:** Microsoft Excel (Pivot Tables, Data Analysis), Azure, GCP, AWS, Jupyter Notebook, VS Code, Streamlit
- **Languages:** Python (NumPy, Pandas, Seaborn, Scikit-learn, NLTK, SpaCy, TensorFlow, LangChain), SQL, UNIX/Linux (shell & systems programming), HTML, CSS, JavaScript, PySpark

LEADERSHIP

SCU Intandesh Club - Cultural Show Director and Bhangra Choreographer

2020-2024

- Directed a 3-hour culture show and Indian celebrations like Diwali and Holi, driving cross-functional collaboration with campus organizations, event coordinators, and marketing teams attracting over 500 attendees and generated \$7000+ in profit.
- Choreographed and led a 12-member Bhangra dance team, performing at various campus events, leading to a 30% increase in our event attendance.